



Complex Fractions

A compound event involves two or more actions. Similarly, a compound or complex fraction has two or more operations.

Definition

A **complex fraction** is a fraction that has a fraction in the numerator and/or denominator.

- **Think About It** A complex fraction can contain numerical fractions, rational expressions, or both.

Examples

$$\frac{\frac{1}{b} + 3}{\frac{2}{b-5}}$$

$$\frac{\frac{1}{4} - x}{\frac{1}{8}}$$

$$\frac{\frac{1}{2}}{\frac{2}{5}}$$

Simplifying a Complex Fraction

To simplify a complex fraction, either multiply the numerator and the denominator of the complex fraction by a common denominator of the fractions within the fraction (the LCD method) or divide the numerator of the complex fraction by the denominator (the division method).

- **Remember** The LCD is the least common denominator of two or more fractions.

EXAMPLE 1

Simplify the expression.

A $\frac{\frac{2}{3}}{\frac{3}{5}}$

SOLUTION

LCD Method

$$\frac{\frac{2}{3}}{\frac{3}{5}} = \frac{\frac{2}{3} \cdot \textcolor{red}{15}}{\frac{3}{5} \cdot \textcolor{red}{15}} = \frac{\frac{2}{\cancel{3}^1} \cdot \overset{5}{15}}{\frac{3}{\cancel{5}_1} \cdot \overset{3}{15}} = \frac{10}{9} = 1\frac{1}{9}$$

Division Method

$$\frac{\frac{2}{3}}{\frac{3}{5}} = \frac{2}{3} \div \frac{3}{5} = \frac{2}{3} \cdot \frac{\textcolor{red}{5}}{\textcolor{red}{3}} = \frac{10}{9} = 1\frac{1}{9}$$

B $\frac{\frac{a}{b}}{\frac{c}{d}}$

SOLUTION

LCD Method

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{\frac{a}{b} \cdot \textcolor{red}{bd}}{\frac{c}{d} \cdot \textcolor{red}{bd}} = \frac{\frac{a}{\cancel{b}^1} \cdot \overset{d}{bd}}{\frac{c}{\cancel{d}_1} \cdot \overset{b}{bd}} = \frac{ad}{bc}$$

Division Method

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{\textcolor{red}{d}}{\textcolor{red}{c}} = \frac{ad}{bc} \quad \blacksquare$$

EXAMPLE 2

Simplify the expression.

A $\frac{\frac{1}{2} + 5}{3 + \frac{1}{4}}$

SOLUTION

$$\begin{aligned}
\frac{\frac{1}{2} + 5}{3 + \frac{1}{4}} &= \frac{\left(\frac{1}{2} + 5\right) \cdot 4}{\left(3 + \frac{1}{4}\right) \cdot 4} && \text{Multiply the numerator and denominator by the LCD: 4.} \\
&= \frac{\frac{1}{2} \cdot 4 + 5 \cdot 4}{3 \cdot 4 + \frac{1}{4} \cdot 4} && \text{Distribute 4.} \\
&= \frac{2 + 20}{12 + 1} && \text{Simplify.} \\
&= \frac{22}{13} && \text{Simplify.} \\
&= 1\frac{9}{13} && \text{Simplify.}
\end{aligned}$$

B $\frac{5 + \frac{1}{2a}}{\frac{3}{2a}}$

SOLUTION

$$\begin{aligned}
\frac{5 + \frac{1}{2a}}{\frac{3}{2a}} &= \left(5 + \frac{1}{2a}\right) \div \frac{3}{2a} && \text{Rewrite as division.} \\
&= \left(\frac{5 \cdot 2a}{2a} + \frac{1}{2a}\right) \div \frac{3}{2a} && \text{Rewrite the first fraction of the dividend using the LCD: } 2a. \\
&= \left(\frac{10a}{2a} + \frac{1}{2a}\right) \div \frac{3}{2a} && \text{Simplify.} \\
&= \left(\frac{10a + 1}{2a}\right) \div \frac{3}{2a} && \text{Simplify.} \\
&= \frac{10a + 1}{2a} \cdot \frac{2a}{3} && \text{Division is the same as multiplication by the reciprocal of the divisor.} \\
&= \frac{10a + 1}{\underset{1}{2a}} \cdot \frac{\overset{1}{2a}}{3} && \text{Divide out the common factor.} \\
&= \frac{10a + 1}{3} && \text{Simplify. } \blacksquare
\end{aligned}$$

Using an LCD to Simplify a Complex Algebraic Fraction

To simplify a complex fraction with variables, multiply the numerator and denominator by the LCD of all fractions that appear in the numerator or denominator. Then simplify. Use this method when there is addition or subtraction in a fraction of the numerator and/or denominator.

EXAMPLE 3

Simplify.

$$\frac{\frac{1}{x-3} + \frac{3}{x}}{\frac{2}{x-3}}$$

SOLUTION

$$\frac{\frac{1}{x-3} + \frac{3}{x}}{\frac{2}{x-3}} = \frac{\frac{1}{x-3} + \frac{3}{x}}{\frac{2}{x-3}} \cdot \frac{x(x-3)}{x(x-3)}$$

Multiply the numerator and denominator by the LCD: $x(x-3)$.

$$= \frac{\frac{1}{x-3} \cdot x(x-3) + \frac{3}{x} \cdot x(x-3)}{\frac{2}{x-3} \cdot x(x-3)}$$

Distribute $x(x-3)$.

$$= \frac{\frac{1}{\cancel{(x-3)}} \cdot \cancel{x} \overset{1}{(x-3)} + \frac{3}{\cancel{x}} \cdot \overset{1}{x} (x-3)}{\frac{2}{\cancel{(x-3)}} \cdot \cancel{x} \overset{1}{(x-3)}}$$

Divide out common factors.

$$= \frac{x + 3(x-3)}{2x}$$

Simplify.

$$= \frac{4x-9}{2x}$$

Simplify. ■